JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 8, 2022
© 2022 BY THE AMERICAN COLLEGE OF CARDIOLOGY FOUNDATION
PUBLISHED BY ELSEVIER

CLINICAL ELECTROPHYSIOLOGY - AI/ML

Use of Wearable Technology and Deep Learning to Improve the Diagnosis of Brugada Syndrome

Shun Liao, MSc,[a] Mahmoud Bokhari, MBBS,[a] Praloy Chakraborty, MD,[a] Adrian Suszko, MSc,[a] Gavin Jones, PhD,[a] Danna Spears, MD,[a] Michael Gollob, MD,[a] Zhaolei Zhang, PhD,[b] Bo Wang, PhD,[a,*] Vijay S. Chauhan, MD[a,*]

ABSTRACT

BACKGROUND The diagnosis of Brugada syndrome by 12-lead electrocardiography (ECG) is challenging because the diagnostic type 1 pattern is often transient.

OBJECTIVES This study sought to improve Brugada syndrome diagnosis by using deep learning (DL) to continuously monitor for Brugada type 1 in 24-hour ambulatory 12-lead ECGs (Holters).

METHODS A convolutional neural network was trained to classify Brugada type 1. The training cohort consisted of 10-second standard/high precordial leads from 12-lead ECGs (n = 1,190) and 12-lead Holters (n = 380) of patients with definite and suspected Brugada syndrome. The performance of the trained model was evaluated in 2 testing cohorts of 10-second standard/high precordial leads from 12-lead ECGs (n = 474) and 12-lead Holters (n = 716).

RESULTS DL achieved a receiver-operating characteristic area under the curve of 0.976 (95% CI: 0.973-0.979) in classifying Brugada type 1 from 12-lead ECGs and 0.975 (95% CI: 0.966-0.983) from 12-lead Holters. Compared with cardiologist reclassification of Brugada type 1, DL had similar performance and produced robust results in experiments evaluating scalability and explainability. When DL was applied to consecutive 10-second, clean ECGs from 24-hour 12-lead Holters, spontaneous Brugada type 1 was detected in 48% of patients with procainamide-induced Brugada syndrome and in 33% with suspected Brugada syndrome. DL detected no Brugada type 1 in healthy control patients.

CONCLUSIONS This novel DL model achieved cardiologist-level accuracy in classifying Brugada type 1. Applying DL to 24-hour 12-lead Holters significantly improved the detection of Brugada type 1 in patients with procainamide-induced and suspected Brugada syndrome. DL analysis of 12-lead Holters may provide a robust, automated screening tool before procainamide challenge to aid in the diagnosis of Brugada syndrome. (J Am Coll Cardiol EP 2022;8:1010–1020)
© 2022 by the American College of Cardiology Foundation.

Brugada syndrome is a rare, inherited arrhythmia afflicting predominantly young male subjects. The majority are asymptomatic, but ventricular tachyarrhythmias can rarely cause syncope and nocturnal sudden death.[1] The diagnosis is based on the presence of a Brugada type 1 pattern on a standard 12-lead electrocardiogram (ECG), which is characterized by coved ST-segment

From the [a]Peter Munk Cardiac Center, Division of Cardiology, Toronto General Hospital, University Health Network, Toronto, Canada; and the [b]Department of Computer Sciences, University of Toronto, Toronto, Canada. *Drs Wang and Chauhan contributed equally to this work and are joint senior authors.
The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received January 13, 2022; revised manuscript received March 28, 2022, accepted May 4, 2022.

ISSN 2405-500X/$36.00 https://doi.org/10.1016/j.jacep.2022.05.003

elevation and T-wave inversion in the standard or high anterior precordial leads.[1,2] However, the sensitivity of a single 12-lead ECG recording in detecting this pattern can be poor because many patients demonstrate marked temporal variations in Brugada type 1.[3] To address this limitation, wearable, ambulatory 12-lead electrocardiograms (ie, Holters) with standard and high precordial leads can improve diagnostic yield of the spontaneous Brugada type 1 ECG pattern by up to 30%,[4,5] but this approach requires manual over-reading of hundreds of 12-lead ECGs over 24 hours, which is neither feasible for population screening nor reliable in the presence of sporadic ECG noises and motion artifacts. Accurate and efficient 12-lead ECG screening of Brugada syndrome is essential to direct early therapeutic intervention and family screening that may prevent sudden death.

In this regard, machine learning (ML), and more specifically deep learning (DL), has been used to automate the classification of complex biomedical signals from ECG recordings,[6] but the utility of DL to classify Brugada syndrome in ambulatory ECGs has not been reported. DL has the advantage of automatically learning features from raw signals without the need for a priori manual feature engineering.[7] We hypothesized that classifying Brugada type 1 in a 24-hour 12-lead Holter recording using DL after systematic noise control will improve the diagnosis of Brugada syndrome. Our objective was to develop a DL model that automated classification of Brugada type 1 and then apply the model to 24-hour 12-lead Holters of patients with definite and suspected Brugada syndrome to detect the Brugada type 1 pattern.

SEE PAGE 1021

METHODS

PATIENT POPULATION. The study included 3 cohorts from our inherited arrhythmia clinic: 1) definite Brugada syndrome; 2) suspected Brugada syndrome; and 3) healthy control patients. Definite Brugada syndrome was defined as a spontaneous or induced (ie, intravenous procainamide 10-15 mg/kg at 50 mg/min, fever, or exercise test) Brugada type 1 ECG pattern in the standard precordial (ie, V_1, V_2 in fourth intercostal space) or high precordial leads (ie, V_1, V_2 in third or second intercostal space).[1,2] Suspected Brugada syndrome included patients with a spontaneous or induced type 2 or type 3 Brugada ECG pattern in the standard/high precordial leads.[1,2] In the absence of this ECG pattern, suspected Brugada syndrome also included patients with a family history of Brugada syndrome/unexplained sudden death, unexplained syncope, or SCN5A pathogenic variant. Healthy control patients had no cardiac symptoms, normal 12-lead ECG/stress tests, and normal echocardiograms. The study was approved by the Institutional Review Board of University Health Network and prospectively enrolled patients provided written informed consent.

ABBREVIATIONS AND ACRONYMS

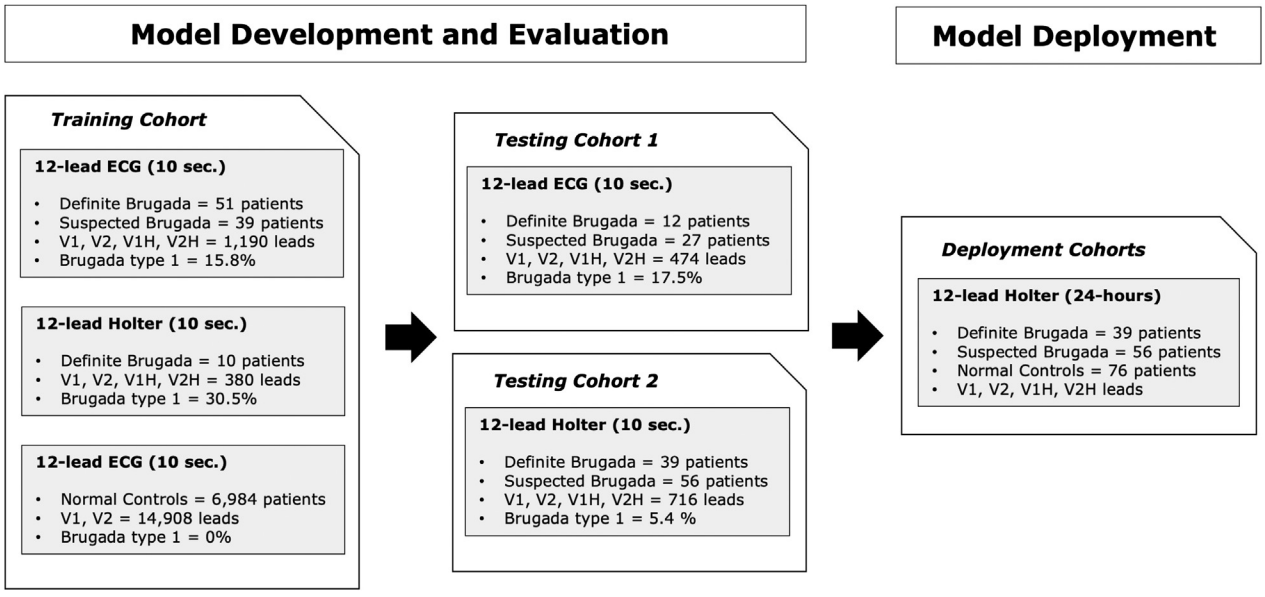
1D = 1-dimensional
AUC = area under curve
DL = deep learning
ML = machine learning
ROC = receiver-operating characteristic

BRUGADA TYPE 1 LABELING OF 12-LEAD ECG AND 12-LEAD HOLTERS. Digital 12-lead ECGs (500 Hz sampling rate) were recorded in patients with definite and suspected Brugada syndrome who were identified from our inherited arrhythmia clinic, whereas digital 12-lead ECGs from healthy control patients were collected from a large public ECG database.[8,9] Digital 12-lead ECGs were also recorded using high precordial leads for those patients undergoing intravenous procainamide challenge. In addition, digital 12-lead Holters (CardioMem CM 3000-12BT, Getemed Inc) were recorded for 24-hours (1,024 Hz sample rate) using standard/high precordial leads in prospectively enrolled patients with definite or suspected Brugada syndrome and healthy control patients from our inherited arrhythmia clinic. The 24-hour 12-lead ECG waveforms were downloaded from the Holters for custom analysis using MATLAB (MathWorks, Inc). For each standard/high precordial lead of the Holter, consecutive 10-second ECG segments were analyzed for baseline wander and high frequency noise as detailed in the Supplemental Methods,[10] and only clean ECG segments were analyzed.

For all 12-lead ECGs and a subset of 12-lead Holters, the standard/high precordial leads were each manually labeled by the consensus of 2 cardiologists as either Brugada type 1 or non-Brugada type 1 (ie, Brugada type 2 or 3, incomplete/variant right bundle branch block, or normal QRST complex) using published criteria.[1,2] In addition, leads V_1 and V_2 from the public database of normal 12-lead ECGs were labeled as non-Brugada type 1.[8,9]

TRAINING AND TESTING COHORT. As shown in Figure 1, a training cohort was constructed by combining the following labeled 10-second ECG data sets: 1) 1,190 standard/high precordial leads from the 12-lead ECGs of 90 patients with definite and suspected Brugada syndrome; 2) 380 standard/high precordial leads from the 12-lead Holters of 10 patients with definite Brugada syndrome, where ten 10-second ECG segments were randomly selected per patient over the 24-hour recording; and 3) 14,908 standard precordial leads from 12-lead ECGs of 6,984 healthy control patients in the public ECG

FIGURE 1 DL Model Development, Evaluation, and Deployment



Deep learning (DL) was developed and evaluated in separate training and testing cohorts composed of standard and high anterior precordial leads from patients with definite Brugada syndrome and those with suspected Brugada syndrome as well as healthy control patients. DL was then deployed in a separate cohort with definite Brugada syndrome, suspected Brugada syndrome, and healthy control patients to classify Brugada type 1. For each cohort, the prevalence of Brugada type 1 among the electrocardiogram (ECG) leads is indicated.

database.[8,9] Together, these 3 data sets introduced sufficient heterogeneity in ECG waveforms to improve training model robustness. Supplemental Table 1 demonstrates the effectiveness of the public ECG database for model training.

The testing cohorts included the following labeled 10-second ECG data sets that did not overlap with the training cohort: 1) 474 standard/high precordial leads from 12-lead ECGs of 39 patients with definite and suspected Brugada syndrome; and 2) 716 standard/high precordial leads from 12-lead Holter of 95 patients with definite and suspected Brugada syndrome, where 2 10-second ECG segments were randomly selected per patient over the 24-hour recording. For both training and testing cohorts, ECG preprocessing was performed as detailed in the Supplemental Methods.

DL MODEL ARCHITECTURE AND EVALUATION. The DL model was designed to input the 10-second ECG recording and then output the probability of Brugada type 1 for each standard/high precordial lead, on a continuous scale from 0 to 1. The model is a 1-dimensional (1D) residual, convolutional, deep neural network, inspired by ResNet-18[11] for image recognition and implemented using PyTorch.[12] The 2D convolution filters in ResNet-18 were replaced by 1D filters to make the architecture more suitable for individual ECG lead processing. As shown in Supplemental Figure 1, the DL architecture is an 18-layer neural network composed of 5 residual convolutional blocks such that each block abstracts the features gradually from the ECG waveform to a higher-level representation. The last layer of our network is a fully connected layer with Softmax function, outputting the probability of Brugada type 1. The proposed architecture was also benchmarked to other DL architectures and traditional supervised ML models as presented in the Supplemental Methods. Details regarding DL model training and data augmentation are provided in the Supplemental Methods, Supplemental Figure 2, and Supplemental Table 1.

BENCHMARKING THE DL MODEL TO CARDIOLOGISTS. The performance of DL in classifying Brugada type 1 was also benchmarked against 2 cardiologists independently performing blinded reclassification of 10-second 12-lead ECGs as Brugada type 1 versus non-Brugada type 1. Reclassification was performed in standard/high precordial leads from a random sample of 25 Brugada type 1 and 25 non-Brugada type 1 12-lead ECGs in the testing cohort. The sensitivity and specificity of Brugada reclassification by the

cardiologists was evaluated relative to the gold standard labels assigned to the testing cohort.

APPLYING THE DL MODEL TO 24-HOUR 12-LEAD HOLTERS. The DL model for classifying Brugada type 1 was then applied to a separate deployment cohort with 24-hour 12-lead Holter recordings that were not used for model training or testing. The 12-lead Holter ECGs were down-sampled and filtered to match training data (3-100 Hz). Only 10-second ECG segments with at least 1 clean standard or high precordial lead were analyzed. Each clean ECG lead was assigned a Brugada type 1 probability by the DL classifier and categorized as Brugada type 1 if this probability exceeded 0.50 in any ECG lead. For each patient, Brugada burden was computed over a 15-minute epoch and defined as the proportion of 10-second ECG segments with Brugada type 1. Over a 24-hour period, Brugada burden was defined as the average of the Brugada burdens in each 15-minute epoch.

STATISTICAL ANALYSIS. Details of the statistical analysis are presented in the Supplemental Methods.

RESULTS

PATIENT CHARACTERISTICS. A total of 129 patients with definite or suspected Brugada syndrome provided 1,664 standard/high precordial lead recordings from their respective 12-lead ECGs. The prevalence of Brugada type 1 in these ECG leads was 16.3%. In addition, a prospectively enrolled cohort of 181 patients wore the 24-hour 12-lead Holter, including: 1) 49 patients with definite Brugada syndrome (age 48 ± 12 years, 88% male); 2) 56 patients with suspected Brugada syndrome (age 43 ± 16 years, 63% male); and 3) 76 healthy control patients (age 38 ± 13 years, 53% male). The clinical characteristics of these patients are presented in Table 1. The definite and suspected Brugada groups provided 1,096 standard/high precordial Holter recordings.

DL PERFORMANCE IN CLASSIFYING BRUGADA TYPE 1 FROM 12-LEAD ECGs IN THE TESTING COHORT. The testing cohort of 12-lead ECGs (ie, testing cohort 1 in Figure 1) comprised 474 standard/high precordial leads from 39 patients with definite and suspected Brugada syndrome, where the prevalence of Brugada type 1 was 17.5%. The performance of DL in classifying Brugada type 1 in this testing cohort is presented in Figure 2A. DL achieved high receiver-operating characteristic (ROC) area under the curve (AUC) of 0.976 (95% CI: 0.973-0.979). The performance of DL in classifying Brugada type 1 was also evaluated using different prediction thresholds. Because the DL model has continuous output, ranging from 0 to 1, the prediction was classified as Brugada type 1 when the DL output was above a threshold, which was based on achieving a predefined sensitivity of 50%, 80%, or 90% in detecting Brugada type 1. The respective specificity, positive predictive value, negative predictive value, F1 score, and accuracy are shown in Table 2. DL had reasonably high specificity for each predefined sensitivity. When compared with other DL architectures and traditional supervised ML models, our DL model's performance was superior based on the ROC AUC as shown in Supplemental Tables 2 and 3.

In the testing cohort of 50 ECGs where Brugada type 1 was reclassified by cardiologists, the interobserver variability of 2 cardiologists relative to the original labels was low based on Cohen's kappa statistic of 0.84 and 0.92, respectively. In the same cohort, the DL model's classification of Brugada type 1 had an ROC AUC of 0.989 (95% CI: 0.984-0.994)

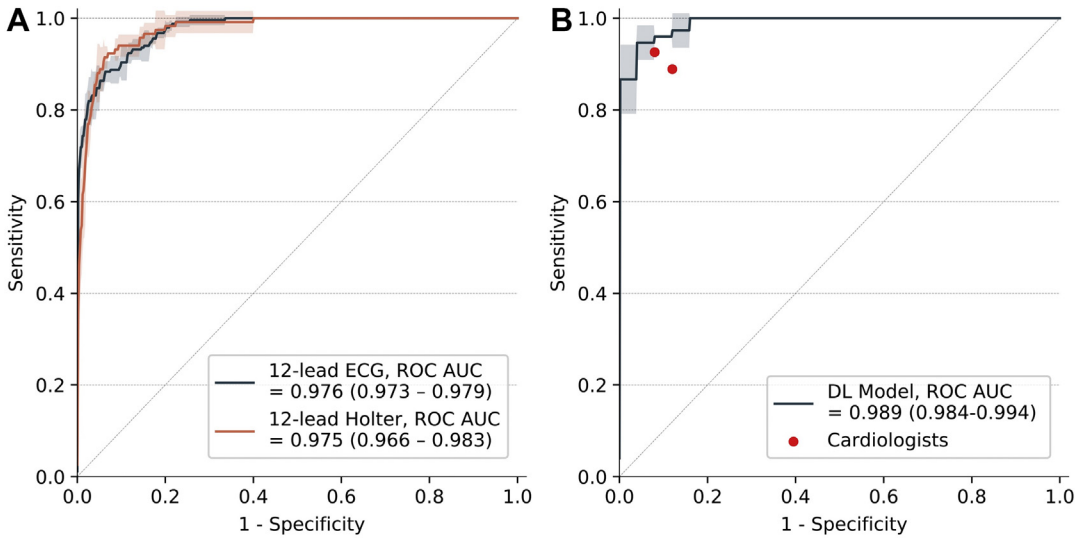
TABLE 1 Characteristics of Patients Wearing 24-Hour 12-Lead Holters

	Definite Brugada Syndrome (n = 49)	Suspected Brugada Syndrome (n = 56)	Healthy Control (n = 76)
Age, y	48 ± 12	43 ± 16	38 ± 13
Male	46 (88)	35 (63)	40 (53)
12-lead ECG pattern			
Brugada type 1	49 (100)	0	0
Spontaneous	24 (49)	0	0
Induced	25 (51)	0	0
Drug	16 (33)	0	0
Fever/exercise testing	9 (18)	0	0
Brugada type 2[a]	0	26 (46)	0
Brugada type 3[a]	0	18 (32)	0
Other[a,b]	0	12 (21)	0
Unexplained aborted cardiac arrest	2 (4)	2 (4)	0
Unexplained syncope	6 (12)	4 (7)	0
Family history of Brugada syndrome	6 (12)	10 (18)	0
Family history of unexplained SCD	8 (16)	7 (13)	0
SCN5A pathogenic variant[c]	7 (16)	2 (13)	–
Inducible VT/VF on EPS[d]	3 (32)	0	–
Right ventricular imaging[e]			
Normal RV size[f]	37 (79)	46 (85)	0
Normal RV function[g]	46 (98)	53 (98)	0
No regional RV abnormalities	45 (96)	51 (94)	0
ICD implanted	8 (16)	2 (4)	0

Values are mean ± SD or n (%). [a]Procainiamide challenge performed in 10 patients (38%) with Brugada type 2, 3 (17%) with Brugada type 3, and 4 (33%) with other. [b]8 patients had family history of Brugada syndrome; 2 had family history of unexplained SCD with Brugada-like ECGs; 1 had unexplained syncope; and 1 had SCN5A pathogenic variant with first-degree relative having Brugada-like ECG. [c]Brugada genotyping was performed in 60 patients. [d]EPS was performed in 18 patients. [e]RV imaging was not available for 4 patients. [f]Abnormal RV size was defined as >100 mL/m^2 (male patients) and >90 mL/m^2 (female patients). [g]Abnormal RV function was defined as RV ejection fraction <45%.

ECG = electrocardiogram; EPS = electrophysiology study; ICD = implantable cardioverter-defibrillator; RV = right ventricular; SCD = sudden cardiac death; VF = ventricular fibrillation; VT = ventricular tachycardia.

FIGURE 2 Performance of DL in Classifying Brugada Type 1 in the Testing Cohort



(A) Receiver-operating characteristic (ROC) curve is shown for DL classification of Brugada type 1 in the testing cohort's 12-lead ECGs and ambulatory 12-lead ECGs (Holters). **(B)** ROC curve is shown for DL applied to a random sample of Brugada type 1 and non-Brugada type 1 12-lead ECGs from the testing cohort, which is compared to the performance of 2 cardiologists reclassifying Brugada type 1 in the same ECG sample. AUC = area under curve; other abbreviations as in Figure 1.

(Figure 2B). The sensitivity and specificity in classifying Brugada type 1 was 96% and 90%, respectively, which was similar to that of the 2 cardiologists reclassifying Brugada type 1 (cardiologist 1: sensitivity 88.9%, specificity 88.0%; cardiologist 2: sensitivity 92.5%, specificity 92.0%) (Figure 2B).

DL PERFORMANCE IN CLASSIFYING BRUGADA TYPE 1 FROM 12-LEAD HOLTERS IN THE TESTING COHORT. The performance of DL in classifying Brugada type 1 from the 12-lead Holters was benchmarked against the 716 labeled, 10-second standard/high precordial ECG segments from the 95 patients with definite or suspected Brugada syndrome in the testing cohort (ie, testing cohort 2 in Figure 1), where the prevalence of Brugada type 1 was 5.4%. The DL model achieved a high ROC AUC of 0.975 (95% CI: 0.966-0.983) and its performance at predefined sensitivities of 50%, 80%, and 90% were excellent as summarized in Figure 2A and Table 2.

ROBUSTNESS OF DL IN CLASSIFYING BRUGADA TYPE 1. The robustness of our DL model was evaluated by assessing variability, scalability, and explainability. Variability was defined by the SD of Brugada type 1 classification performance. As shown

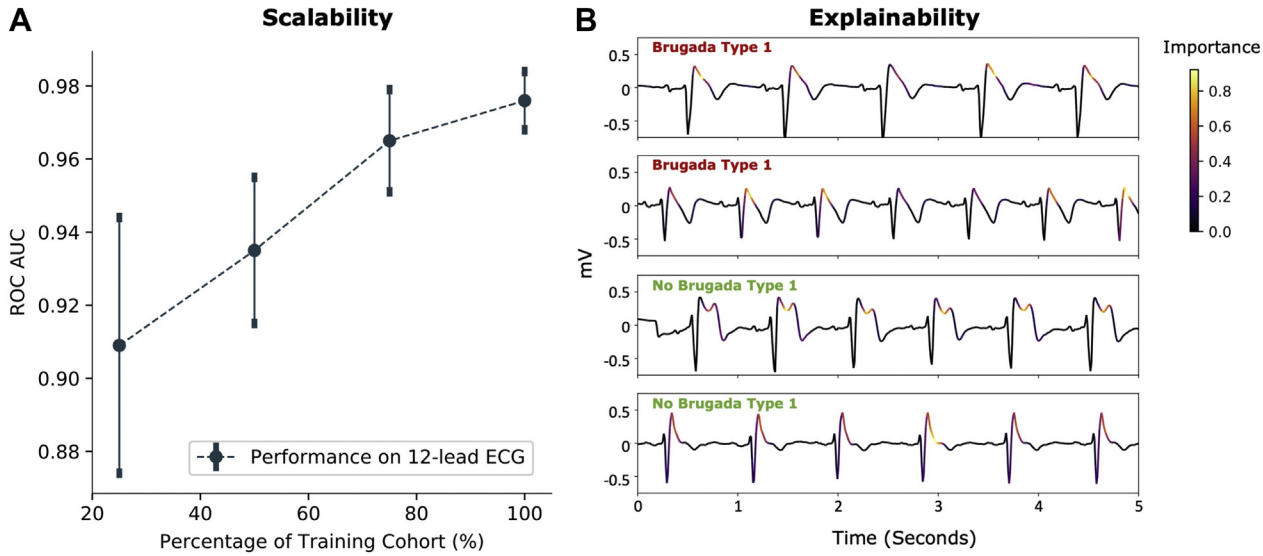
TABLE 2 DL Performance in Classifying Brugada Type 1 in the Testing Cohort

ECG Data	Brugada Prevalence[a] % (n)	ROC AUC	Sensitivity	Specificity	NPV	PPV	F1 Score	Accuracy
12-lead ECG	17.5 (474)	0.976 (0.973-0.979)	50	100.0 (100.0-100.0)	90.5 (90.5-90.5)	100.0 (100.0-100.0)	67.2 (67.2-67.2)	91.4 (91.4-91.4)
			80	97.2 (95.0-99.4)	96.0 (95.9-96.0)	86.2 (76.9-95.4)	83.3 (78.9-87.7)	94.3 (92.5-96.1)
			90	90.5 (87.4-93.7)	97.8 (97.7-97.9)	67.2 (59.7-74.7)	77.0 (72.1-81.9)	90.5 (87.9-93.1)
12-lead Holter	5.4 (716)	0.975 (0.966-0.983)	50	99.3 (98.7-99.9)	97.3 (97.2-97.3)	81.7 (67.7-95.6)	62.9 (58.8-67.0)	96.7 (96.1-97.3)
			80	96.8 (95.7-98.0)	98.9 (98.9-99.0)	60.3 (51.2-69.4)	69.4 (63.5-75.4)	96.0 (95.0-97.1)
			90	94.2 (92.3-96.1)	99.5 (99.5-99.5)	48.2 (39.6-56.7)	63.2 (55.9-70.5)	94.1 (92.3-95.9)

Values are % (n) or HR (95% CI). [a]Prevalence among standard and high precordial leads.

AUC = area under curve; DL = deep learning; ECG = electrocardiography; Holter = ambulatory ECG; NPV = negative predictive value; PPV = positive predictive value; ROC = receiver-operating characteristic.

FIGURE 3 Robustness of DL Model



(A) The scalability of DL is evaluated by plotting its ROC AUC as a function of the training cohort size. **(B)** The explainability of DL is illustrated using importance plots generated by Grad-CAM for 2 patients with Brugada type 1 and 2 patients without Brugada type 1. In the classification of Brugada type 1 by DL, the highest importance in the ECG waveform is assigned to the ST segments of each patient. Abbreviations as in Figures 1 and 2.

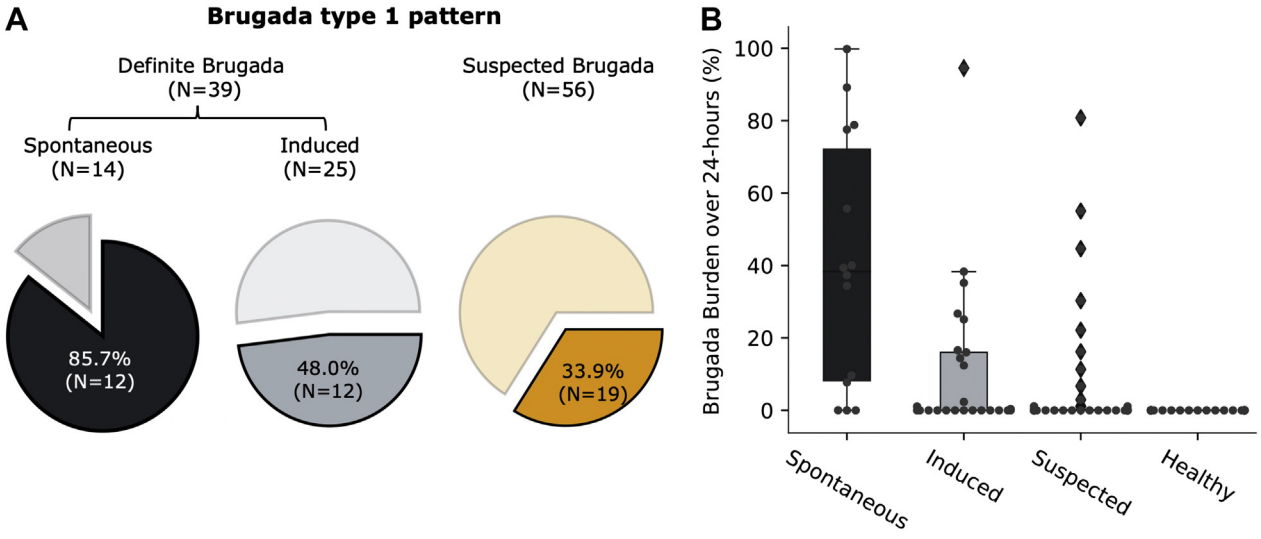
in Table 2, the variance in AUC was <1% for different neural network initializations, which indicated low model variability and repeatable, robust performance. Regarding scalability, the performance of DL was evaluated using smaller training cohorts to determine whether the training cohort was adequately sized. The DL model was trained with 25%, 50%, and 75% of the original training cohort and then evaluated using the same testing cohort. As shown in Figure 3A, an increase from 75% to 100% was associated with a marginal change in ROC AUC from 0.965 to 0.976 in the testing cohort, suggesting that our training cohort was adequately sized. In the case of explainability, the DL model's output was explained using a gradient-based method (Grad-CAM)[13] that computes the relative importance of QRST waveform components in classifying Brugada type 1 (see Supplemental Methods). Figure 3B shows 2 patients classified as Brugada type 1 by DL where Grad-CAM assigned the highest importance to the ST segments in explaining this classification. The explanation for the model's output was similar to clinical interpretation; thereby further supporting the model's robustness.

DL DETECTION OF BRUGADA TYPE 1 FROM 24 HOUR 12-LEAD HOLTER IN THE DEPLOYMENT COHORT. From the 24-hour 12-lead Holter recording, the proportion of 10-second 12-lead ECG segments with at least 1 clean ECG lead was similar among patients with definite Brugada syndrome (51.3%), those with suspected Brugada syndrome (50.3%), and healthy control patients (51.1%) ($P = 0.521$). Supplemental Figure 3 illustrates examples of low- and high-frequency noisy 10-second ECG segments, which would be excluded. As shown in Supplemental Figure 4, the majority of clean segments were noted in the late evening and night time (ie, 10 PM to 7 AM).

During the 24-hour 12-lead Holter recordings, DL identified 43 patients with at least one 15-minute epoch containing >1% Brugada burden, of whom 24 patients (55.8%) were definite Brugada syndrome and 19 (33.9%) were suspected Brugada syndrome. None of the healthy control patients had more than 1% Brugada burden in any 15-minute epoch. Among patients with definite Brugada syndrome, DL identified Brugada type 1 in 85.7% with spontaneous Brugada type 1 and 48.0% with induced-Brugada type 1 (Figure 4A). In the 16 patients with procainamide-induced Brugada type 1, DL identified Brugada type 1 in 8 patients (50%), while in those with a negative procainamide challenge ($n = 23$), DL identified Brugada type 1 in 2 patients (8.7%). Brugada burden over 24 hours was greater in definite Brugada syndrome compared with suspected Brugada syndrome (21.8% vs 4.9%; $P < 0.001$) as shown in Figure 4B. Among patients with definite Brugada syndrome, the

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 8, 2022
AUGUST 2022:1010–1020

FIGURE 4 DL Classification of Brugada Type 1 in 24-Hour 12-Lead Holters



(A) Pie charts show the prevalence of Brugada type 1 based on DL classification in patients with definite (ie, spontaneous or induced) and suspected Brugada syndrome. **(B)** Brugada burden over 24 hours defined by DL is plotted for the same group of patients as well as healthy control patients. Abbreviations as in Figures 1 and 2.

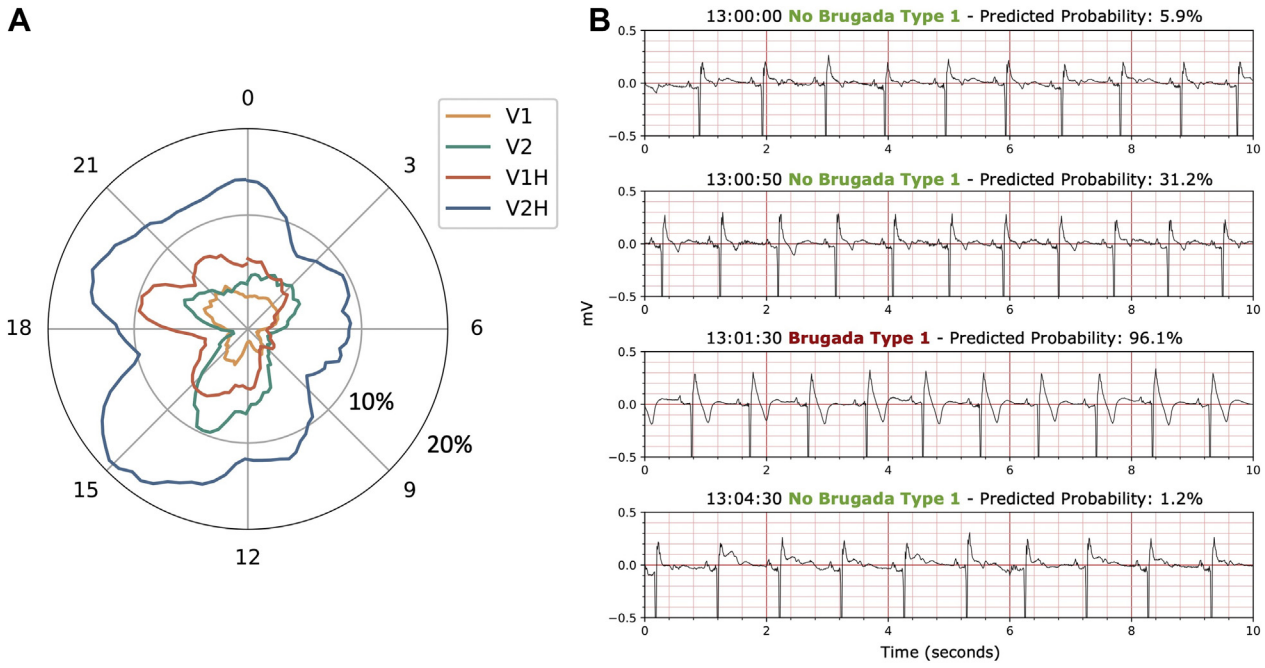
Brugada burden over 24 hours was greater in those with spontaneous Brugada type 1 compared to those with induced Brugada type 1 (40.7% vs 11.3%; $P < 0.001$).

Regarding the diagnostic yield of each ECG lead, the standard and high precordial leads identified 38 (40.4%) and 25 (26.6%) patients, respectively, with more than 1% Brugada burden over 24 hours. Figure 5A illustrates the 24-hour temporal Brugada burden for standard and high precordial leads averaged over all patients with definite and those with suspected Brugada syndrome. The left parasternal region (ie, V2H) disclosed the greatest Brugada burden among all leads, which ranged from 12% to 15% over 24 hours. Brugada burden had a bimodal temporal distribution with peaks at 1:00 PM to 4:00 PM and 7:00 PM to midnight corresponding in many patients to their postprandial state. The postprandial temporal variability in Brugada burden is illustrated in Figure 5B for 1 patient with suspected Brugada syndrome where Brugada type 1 developed over 90 seconds at 1:00 PM shortly after lunch, but then resolving a few minutes later. This patient had a negative procainamide challenge and would not have been classified as definite Brugada syndrome without the 12-lead Holter disclosure. The temporal variability in Brugada burden over 24 hours is shown in Figure 6 for all 43 patients, who are arranged in this Figure according to hierarchical cluster analysis of their temporal burden (see the Supplemental Methods).

DISCUSSION

This study developed a novel DL model to automate classification of the Brugada type 1 pattern in ambulatory 12-lead ECG recordings with careful attention to noise control. The major findings are as follows: 1) DL trained on standard/high precordial leads from 12-lead ECGs and 12-lead Holters showed excellent discriminatory performance (ROC AUC >0.97, accuracy >90%) for Brugada type 1 in a testing cohort of 12-lead ECGs and 12-lead Holters; 2) In a subset of standard/high precordial leads from 12-lead ECGs, DL's performance was similar to that of cardiologists reclassifying Brugada type 1; and 3) one-half of patients with induced Brugada type 1 and one-third of patients with suspected Brugada syndrome had a spontaneous Brugada type 1 ECG detected by DL in their 24-hour 12-lead Holters. Thus, our DL model increased the diagnostic yield for Brugada syndrome compared to traditional 12-lead ECGs by providing a reliable, automated means of disclosing spontaneous Brugada type 1

FIGURE 5 Diagnostic Yield of Standard and High Precordial Leads Over 24 Hours Using DL



(A) Brugada burden assessed by DL is plotted for each standard and high precordial lead of the 12-lead Holter during the 24-hour recording. Brugada burden was averaged from all patients to determine the diagnostic yield of each lead. The concentric circles indicate a Brugada burden of 10% and 20%. **(B)** DL analysis of 12-lead Holter in a patient with suspected Brugada syndrome revealed Brugada type 1 in lead V2H lasting 90 seconds at 1:00 PM shortly after lunch and resolving a few minutes later. This patient had a negative procainamide challenge. Abbreviations as in Figures 1 and 2.

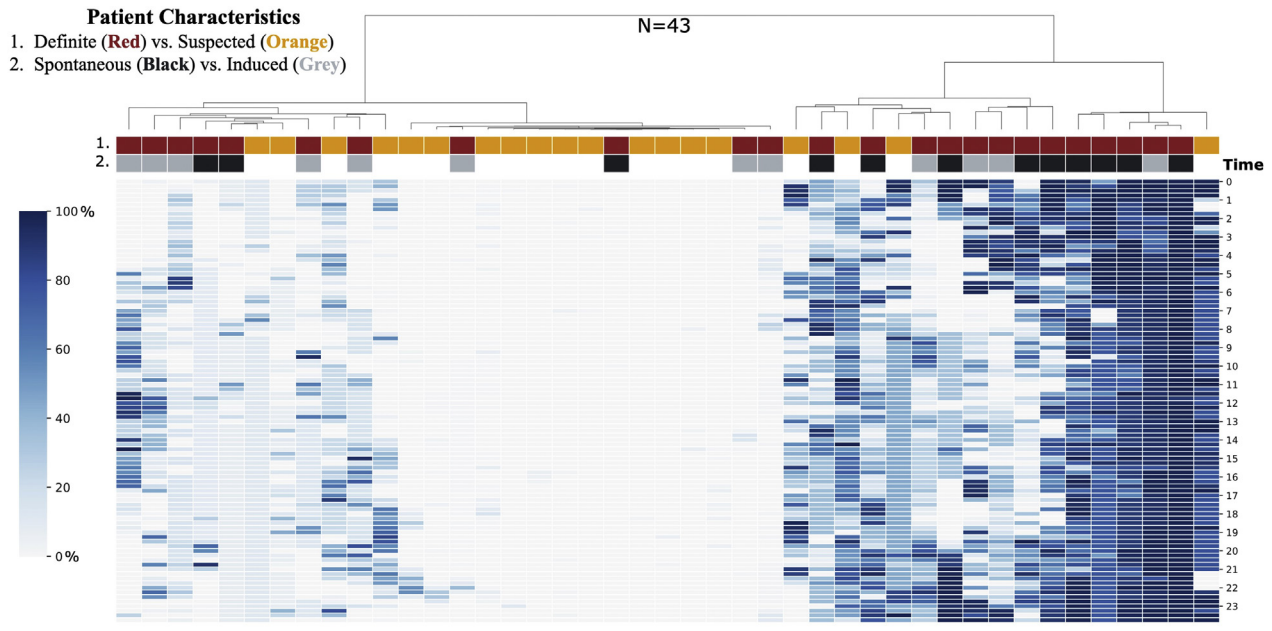
during 24-hour ambulation. Our explainability analysis using Grad-CAM[13] suggested that the ST-segment is important in the classification of Brugada type 1 using DL, which is similar to clinical interpretation.

DL models have previously been developed to classify arrhythmias[6] and detect disease-specific ECG features[14] from 12-lead ECGs and short duration single-lead recordings. The application of DL in 24-hour Holter recordings to track beat-to-beat ECG waveform features has not been previously reported to our knowledge. Ambulatory monitoring with 12-lead disclosure may capture diagnostic ECG patterns that are unmasked by changing heart rate or autonomic tone during meals, activity, or sleep. Our study demonstrates that combining wearable ECG monitoring with DL-based detection is feasible and incremental over standard of care in diagnosing Brugada syndrome. This approach also has potential to improve ECG-based diagnosis and phenotyping for other rare inherited arrhythmia disorders, such as congenital long QT syndrome, where diagnostic ECG features are often transient.

COMPARISON WITH PRIOR STUDIES AND CLINICAL IMPLICATIONS. The utility of 12-lead Holter monitoring to diagnose Brugada syndrome has been reported using manual over-reading at 5-minute or longer intervals, but without prespecified noise control. In 54 patients with definite Brugada syndrome, Gray et al[5] demonstrated a spontaneous Brugada type 1 pattern in the standard/high precordial leads at least once during the 24-hour 12-lead Holter recording in 34% of those with drug-induced Brugada syndrome. Cerrato et al[4] also demonstrated spontaneous Brugada type 1 in 20% of 176 patients with ajmaline- or flecainide-induced Brugada syndrome using 24-hour 12-lead Holters with standard/high precordial leads. With our DL model, the detection of spontaneous Brugada type 1 among patients with induced Brugada syndrome was significantly higher at 48% because automated ECG waveform analysis was performed in consecutive 10-second intervals. This high temporal resolution will enhance the detection of transient Brugada type 1 lasting even a few minutes (Figure 5), which may arise from sympathovagal surges after meals and during sleep.[15] DL identified the greatest

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 8, 2022
AUGUST 2022:1010–1020

FIGURE 6 Temporal Distribution of Brugada Burden Over 24 Hours Using DL



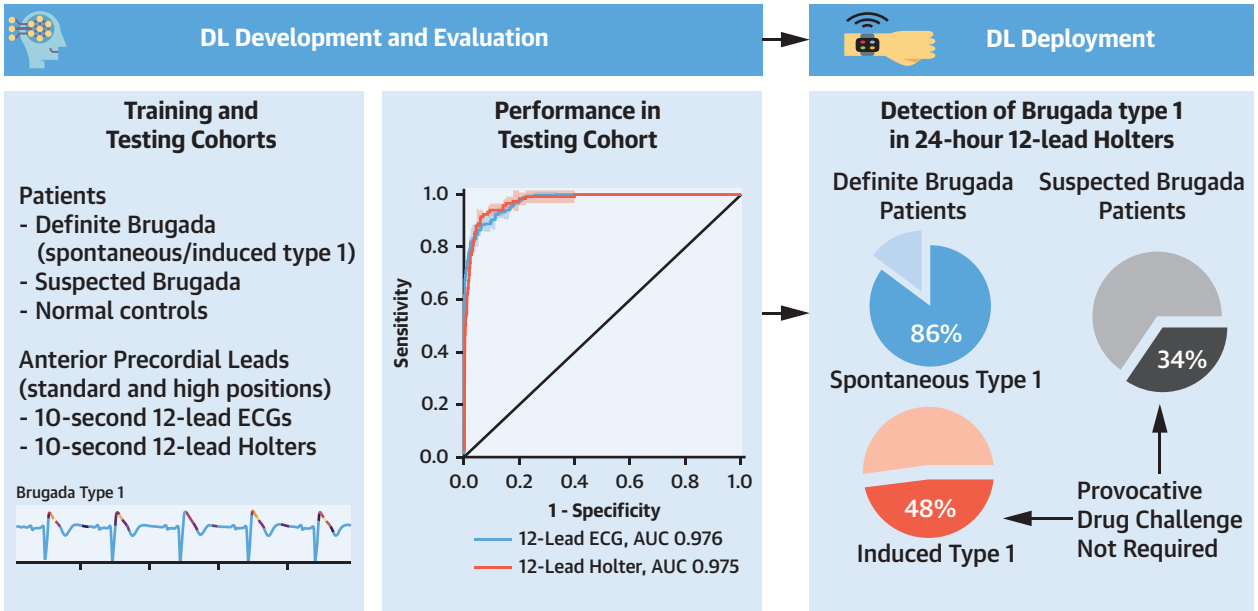
Brugada burden is plotted in consecutive 15-minute epochs over 24 hours for each patient with >1% Brugada burden. The 43 patients shown are ordered based on hierarchical cluster analysis of their temporal Brugada burden over 24 hours. At least 4 phenogroups are evident, 1 with persistently high Brugada burden, 1 with persistently low Brugada burden, and 2 with marked temporal variability in Brugada burden over 24 hours. DL = deep learning.

Brugada burden in the afternoon and evening rather than at night with the high precordial leads in the left parasternal region (ie, V2H) providing the most disclosure. Despite our exclusion of noisy ECG segments predominantly during daytime hours, our findings are similar to the aforementioned 12-lead Holter studies using manual classification.[4,5]

Our DL model also detected spontaneous Brugada type 1 in one-third of patients with suspected Brugada syndrome as well as 8.7% of patients with negative procainamide-challenge, which has not been previously reported. The majority of patients with suspected Brugada syndrome had a spontaneous or procainamide-induced type 2 or 3 ECG pattern. Procainamide challenge was not performed in 69% of these patients because their clinical presentation and demographics (ie, female, asymptomatic, non-Asian) suggested a low pretest probability for Brugada syndrome. This highlights the logistic challenges in performing more routine provocative drug testing to complete diagnostic work-up in suspected Brugada syndrome. The sensitivity and reproducibility of pharmacologic testing may vary depending on the agent used, basal autonomic tone, and genotype.[16] The sensitivity of procainamide is notably less than other more potent sodium channel blockers, such as ajmaline or flecainide, which may account for the false-negative procainamide challenges in our study.[16] DL analysis of 12-lead Holters represents a more practical screening tool for Brugada syndrome, and in some patients it may be more sensitive than procainamide. As a screening tool, a low false-positive rate is desirable, and this was confirmed among our healthy control patients in whom no Brugada burden was detected by DL. Our findings suggest that patients with suspected Brugada syndrome should be screened with 12-lead Holters and DL-based Brugada type 1 detection before considering more invasive procainamide testing (**Central Illustration**). Further discussion regarding ML applications in Brugada syndrome is provided in the Supplemental Appendix.

STUDY LIMITATIONS. The limitations of our study are provided in the Supplemental Appendix.

CENTRAL ILLUSTRATION **Deep Learning Detection of Brugada Type 1 From Wearables**



Liao S, et al. J Am Coll Cardiol EP. 2022;8(8):1010–1020.

The deep learning (DL) model was developed and evaluated in training and testing cohorts, respectively, consisting of standard/high anterior precordial leads from 10-second 12-lead electrocardiographs (ECGs) and ambulatory 12-lead ECGs (Holters) of patients with definite Brugada syndrome (spontaneous and induced type 1), those with suspected Brugada syndrome, and healthy control patients. When applied to a separate cohort of 24-hour 12-lead Holters from patients with definite and those with suspected Brugada syndrome, our DL model detected Brugada type 1 in 48% of patients with induced Brugada syndrome and 34% of those with suspected Brugada syndrome. Screening with DL and wearables would avoid the need for provocative drug testing in these patients. AUC = area under the curve; DL = deep learning.

CONCLUSIONS

Our DL model trained on 12-lead ECGs and 12-lead Holters provided cardiologist-level accuracy in classifying the Brugada type 1 pattern in patients with definite or suspected Brugada syndrome. When applied to 24-hour, 12-lead Holters, DL detected spontaneous Brugada type 1 in one-half of patients with drug-induced Brugada syndrome and one-third of those with suspected Brugada syndrome, who would otherwise remain undiagnosed. DL may provide a robust, automated screening tool before procainamide challenge to aid in the diagnosis and phenotyping of Brugada syndrome. By extension, wearable, ambulatory ECG monitoring combined with automated DL-based waveform analysis may be scalable to population-based screening of other inherited arrhythmia disorders where diagnostic ECG features are transient.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Bo Wang or Dr Vijay Chauhan, Peter Munk Cardiac Center, Division of Cardiology, Toronto General Hospital, Gerrard Wing 3-522, 150 Gerrard Street West, Toronto, Ontario M5G 2C4, Canada. E-mail: bo.wang@uhnresearch.ca OR vijay.chauhan@uhn.ca.

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 8, 2022
AUGUST 2022:1010–1020

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: The diagnostic yield for Brugada syndrome using a 12-lead ECG is poor because of the transient nature of the Brugada type 1 pattern. Our DL model trained on standard/high anterior precordial leads from 12-lead ECGs and 12-lead Holters among healthy control patients, patients with definite Brugada syndrome, and those with suspected Brugada syndrome achieved cardiologist-level accuracy in classifying the Brugada type 1 pattern. When DL was applied to 24-hour 12-lead Holters, spontaneous Brugada type 1 was detected in one-half of patients with procainamide-induced Brugada syndrome and in one-third of patients with suspected Brugada syndrome, who would otherwise remain undiagnosed.

TRANSLATIONAL OUTLOOK: DL analysis of 24-hour 12-lead Holters may provide a practical, automated, noninvasive screening tool to diagnose Brugada syndrome before procainamide challenge. The broader translation of our findings is that wearable ECG monitoring combined with automated DL-based ECG waveform analysis may be scalable to population-based screening of other inherited arrhythmia disorders where diagnostic ECG features are transient.

REFERENCES

1. Brugada J, Campuzano O, Arbelo E, Sarquella-Brugada G, Brugada R. Present status of Brugada syndrome: JACC State-of-the-Art Review. *J Am Coll Cardiol*. 2018;72(9):1046–1059.

2. Priori SG, Wilde AA, Horie M, et al. HRS/EHRA/APHRS expert consensus statement on the diagnosis and management of patients with inherited primary arrhythmia syndromes: document endorsed by HRS, EHRA, and APHRS in May 2013 and by ACCF, AHA, PACES, and AEPC in June 2013. *Heart Rhythm*. 2013;10(12):1932–1963.

3. Veltmann C, Schimpf R, Echternach C, et al. A prospective study on spontaneous fluctuations between diagnostic and non-diagnostic ECGs in Brugada syndrome: implications for correct phenotyping and risk stratification. *Eur Heart J*. 2006;27(21):2544-2552.

4. Cerrato N, Giustetto C, Gribaudo E, et al. Prevalence of type 1 Brugada electrocardiographic pattern evaluated by twelve-lead twenty-four-hour Holter monitoring. *Am J Cardiol*. 2015;115(1):52–56.

5. Gray B, Kirby A, Kabunga P, et al. Twelve-lead ambulatory electrocardiographic monitoring in Brugada syndrome: potential diagnostic and prognostic implications. *Heart Rhythm*. 2017;14(6):866-874.

6. Hannun AY, Rajpurkar P, Haghpanahi M, et al. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nat Med*. 2019;25(1):65–69.

7. LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature*. 2015;521(7553):436–444.

8. Wagner P, Strodthoff N, Bousseljot RD, et al. PTB-XL, a large publicly available electrocardiography dataset. *Sci Data*. 2020;7(1):154.

9. Goldberger AL, Amaral LA, Glass L, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation*. 2000;101(23):E215–E220.

10. Satija U, Ramkumar B, Manikandan MS. A simple method for detection and classification of ECG noises for wearable ECG monitoring devices. In: *2015 2nd International Conference on Signal Processing and Integrated Networks*. IEEE; 2015:164–169.

11. He K, Zhang X, Ren S, Sun J. Identify mapping in deep residual networks. In: Leibe B, Matas J, Sebe N, Welling M, eds. *Computer Vision—ECCV 2016. Lecture Notes in Computer Science 9908*. Springer; 2016:630–645.

12. Paszke A, Gross S, Massa F, et al. Pytorch: an imperative style, high-performance deep learning library. *Adv Neural Info Process*. 2019;32:8026–8037.

13. Selvaraju R, Cogswell M, Das A, et al. Grad-CAM: visual explanations from deep networks via gradient-based localization. In: *2017 IEEE International Conference on Computer Vision*. IEEE; 2017:618–626.

14. van de Leur RR, Taha K, Bos MN, et al. Discovering and visualizing disease-specific electrocardiogram features using deep learning: proof-of-concept in phospholamban gene mutation carriers. *Circ Arrhythm Electrophysiol*. 2021;14(2):e009056.

15. Mizumaki K, Fujiki A, Tsuneda T, et al. Vagal activity modulates spontaneous augmentation of ST elevation in the daily life of patients with Brugada syndrome. *J Cardiovasc Electrophysiol*. 2004;15(6):667–673.

16. Therasse D, Sacher F, Petit B, et al. Sodium-channel blocker challenge in the familial screening of Brugada syndrome: safety and predictors of positivity. *Heart Rhythm*. 2017;14(10), 1442-1148.

KEY WORDS Brugada syndrome, diagnosis, Holter ECG, machine learning

APPENDIX For supplemental methods, discussion, references, figures, and tables, please see the online version of this paper.